

CSE 370– Database Systems

QUIZ 2

Question 1:

Refer to the table below to answer the following questions:

Student		
ID	Name	Email
23101235	Abir	Abir@gmail.com
20101239	Rafi	Rafi@gmail.com
21101248	Samia	Samia@gmail.com
22101984	Sifat	Sifat@gmail.com
20101155	Shafi	Shafi@gmail.com
24101665	Rafi	Rafi@gmail.com

Grade		
Student ID	Course Code	Grade
23101235	CSE321	3.70
20101239	CSE110	4.00
21101248	CSE110	3.00
20101155	CSE423	2.70
22101984	CSE321	1.30
23101235	CSE370	0.00

Course		
ID	Course Title	Department
CSE370	Database System	CSE
CSE421	Computer Network	CSE
CSE110	Programming 1	CSE
CSE423	Computer Graphics	CSE
CSE321	Operating System	CSE

(i) If <'21301153', 370.0, '3'> is inserted in the Grade table, identify all constraints which will be violated by this operation and briefly explain your answer. [4 Marks]

(ii) If you delete the course with ID = 'CSE423' from the Course table, identify all constraints which will be violated by this operation and explain how you will solve it. [2 Marks]

Solution:

b(i).

"21301153": Referential integrity violation, as it does not exist in the referenced table.

370.0: Violates both referential integrity (does not exist in the referenced table) and domain constraint (should be a string, not an integer).

"3": Violates the domain constraint, as it should be a decimal value, not a string, and the format is also incorrect.

b(ii). A referential integrity violation will occur when attempting to delete <"CSE423", "Computer Graphics", "CSE"> from the Course table. To resolve this, any of cascade, set null, or restrict can be used.

Question 2:

Select whether it is DDL or DML, and explain briefly:

You need to modify the "Products" table's primary key to include both "product_id" and "supplier_id" columns. [2 Marks]

Answer: DDL

Explanation: Modifying the primary key of the "Products" table changes the table structure, as primary keys define the uniqueness of records and enforce constraints. Since this involves altering the schema rather than manipulating the data itself, it falls under **Data Definition Language (DDL)**.

Question 3:

In an e-commerce platform, the organization decides to implement partitioning on the "Orders" table based on the geographical region. However, users querying the "Orders" table are unaware of this partitioning, and the queries are still executed normally, without requiring any modification in the user interface.

Identify the type of data independence (Logical or Physical) and explain. [2 Marks]

Answer: Physical Data Independence

Explanation: Physical data independence means that changes in the physical storage of the database do not affect how users or applications interact with the data. In this case, partitioning the "Orders" table based on geographical regions changes the way data is stored and accessed internally, but users continue querying the table as usual. Since this modification does not require changes in queries or application logic, it demonstrates physical data independence.

Question 1:

Refer to the table below to answer the following questions:

Player				Tournament				Game		
PID	Name	email	age	TournamentID	PID	GameID	Participants	GameID	Name	genre
101	Alice	alice@gmail.com	25	T1	101	G104	2000	G101	Chess	Strategy
102	Bob	bob@yahoo.com	22	T2	102	G102	1500	G102	Tetris	Puzzle
103	Charlie	charlie@gmail.com	30	T3	101	G103	2500	G103	Call of Duty	FPS
104	David	david@hotmail.com	27	T4	104	G105	3000	G104	Final Fantasy	RPG
105	Eve	eve@outlook.com	24	T2	105	G103	1800	G105	Street Fighter	Fighting

(i) If <'T3',201, NULL, 237.5> is inserted into the Tournament table, identify all constraints which will be violated by this operation and briefly explain your answer. [4 Marks]

(ii) Modify the GameID of Chess to 'G104' and the genre to NULL from the Game Table, identify all constraints which will be violated by this operation and explain how you will solve it. [2 Marks]

Solution:

b. i. **PID 201:** Referential Integrity, does not exist in the referenced table.

237.5: Domain Constraint, as the value of the participants does not match the format. **NULL:** GameID null will violate Entity Integrity since Pk cannot be null. It will also violate Referential since the foreign key is part of its primary key.

b. ii. Key constraint, 'G104' already exists, so the duplicate key will cause a violation. Declare GameID as primary key during table creation to prevent violation.

Question 2:

Select whether it is DDL or DML, and explain briefly:

A database developer needs to modify a foreign key constraint on the "Orders" table to link it with the "Products" table. [2 Marks]

Answer: DDL

Explanation: This operation involves modifying the underlying storage structure of a table, which affects how the data is stored rather than manipulating the data itself. Since it deals with the structure of the database, it falls under **Data Definition Language (DDL)**

Question 3:

In a hospital management system, the IT team changes the underlying storage structure of the "Patient_Records" table to reduce disk space usage without informing the end users. The users still interact with the same "Patient_Records" table through the same interface.

Identify the type of data independence (Logical or Physical) and explain. [2 Marks]

Answer: Physical Data Independence

Explanation: Physical data independence refers to the ability to change the physical storage structure of the database without affecting how users or applications interact with the data. In this scenario, the IT team modifies the underlying storage structure of the "Patient_Records" table to optimize disk space usage, but users continue accessing the table as usual. This means that the changes occur at the physical level without impacting the logical structure or application layer, demonstrating physical data independence.